

Burnup Calculation Using POD-Based Neutron Spectrum Reconstruction: Application to High-Temperature Gas-Cooled Reactor Core Analysis

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ABSTRACT

We developed a regression model to reconstruct the multi-group neutron flux energy spectrum within a fuel compact for any burnup and operating conditions, given multiple input parameters such as U-235 enrichment and fuel temperature, targeting the high-temperature gas-cooled reactor HTTR. To achieve this, a regression function was created using dimensional reduction with the proper orthogonal decomposition and the linear least-square method. Using the neutron energy spectrum obtained with this regression function through nuclear fuel burnup, we calculated the nuclide inventories and evaluated their errors in 120 random conditions. The results show that the nuclide inventories can be reproduced with good accuracy within few-percent error.

Keywords: Burnup calculation, Proper orthogonal decomposition, Regression model, Nuclide inventory, HTTR

1. INTRODUCTION

Currently, burnup calculations in nuclear reactors are primarily performed using two approaches. The first approach involves solving the neutron transport equation to calculate reaction rates, as implemented in codes such as Serpent. Although this method provides highly accurate results, it requires substantial computational resources and time. The second approach, employed in codes such as ORIGEN, uses pre-generated one-group cross-section libraries for specific fuels and operating conditions. This method offers fast computation but is limited by the finite nature of the cross-section library, which can lead to reduced accuracy under conditions not represented in the library.

To address these limitations, Watanabe et al. proposed a method utilizing the proper orthogonal decomposition (POD) on the snapshot matrix of the neutron energy spectrum to create a linear regression model [1]. While Watanabe's study focused on light water reactors, similar investigations were conducted for fast reactors [2]. Building upon these preceding works, the present research created a regression model for the multi-group neutron energy spectrum targeting the high-temperature gas-cooled reactor HTTR. The objective of the present study is to develop a fast surrogate model for neutron energy spectrum reconstruction intended for use in burnup calculations under prescribed irradiation conditions, similar to ORIGEN-type codes. From this perspective, the unit fuel compact-level model is considered sufficient for demonstrating the feasibility and performance of the proposed POD-based approach. The present framework is not intended to directly resolve lattice- or core-level neutron spectra that explicitly account for spectral interference effect.

In the present model, U-235 enrichment, fuel compact pitch, TRISO packing fraction, linear power, fuel temperature, and burnup is used as input parameters. Here, TRISO packing fraction is a particular parameter to HTTR. These parameters are selected to enable fast burnup calculations for HTTR fuel compacts while accounting for variations in design conditions within a realistic design

space, rather than being restricted to a single fixed design. Using this model, we confirmed that it enables fast burnup calculations for fuel compacts of HTTR.

2. METHOD

In this study, we first solved the multi-group neutron transport equation to calculate the neutron energy spectrum for specific parameters during nuclear fuel burnup. To reconstruct this spectrum, we employed POD and the least-square method to create a polynomial regression model.

2.1 Preparing Snapshot Matrix and Applying POD

First, we obtained the n_g -group neutron energy spectrum by solving the neutron transport equation under various conditions and obtained a snapshot matrix which is constructed by arranging horizontally column vectors of the neutron energy spectrum calculated for each burnup condition.

Burnup calculations were performed and the n_g -groups neutron flux were obtained at n_b burnup points, resulting in a snapshot matrix with n_g rows and n_b columns. Furthermore, since the number of training data having different conditions was n_p , the number of columns in the snapshot matrix subjected to the singular value decomposition became $n_r = n_p \times n_b$. Here, n_r is greater than n_g .

Thus, a snapshot matrix $\mathbf{F}$ formed by arranging n_r pieces of training data for the n_g -group neutron energy spectrum of the side by side is as follows:

$$\mathbf{F} = (\boldsymbol{\phi}_1 \boldsymbol{\phi}_2 \cdots \boldsymbol{\phi}_{n_r}). \quad (1)$$

Here, $\boldsymbol{\phi}_i$ is the n_g -group neutron energy spectrum.

The singular value decomposition was applied to this snapshot matrix to obtain an orthonormal set (a set of POD vectors). Here, the singular value decomposition is the operation of decomposing an $n_g \times n_r$ snapshot matrix into an $n_g \times n_g$ diagonal matrix of singular values, an $n_g \times n_g$ matrix of left singular vectors, and an $n_r \times n_r$ matrix of right singular vectors,

$$\mathbf{F} = \mathbf{U}(\boldsymbol{\Sigma} \quad \mathbf{0})\mathbf{V}^T, \quad (2)$$

where

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_{n_g} \end{pmatrix}. \quad (3)$$

Here, the singular values $\sigma_1, \cdots, \sigma_{n_g}$ are arranged in descending order, and

$$\boldsymbol{\Sigma}\mathbf{V}^T = \begin{pmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_{n_g} \end{pmatrix} (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_{n_r})^T = (\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_{n_r}). \quad (4)$$

As can be seen from this, by obtaining the product of $\boldsymbol{\Sigma}$ and $\mathbf{V}^T$ as n_r expansion coefficient vectors $\mathbf{a}_i$ (position in a POD space), it becomes clear that the first entry of expansion coefficient vector obtained by multiplying σ_1 by the right singular vectors $\mathbf{V}^T$ has the highest importance, while the importance of subsequent rows decreases.

Therefore, when reconstructing neutron spectrum as a linear sum of $\mathbf{u}_i$ for computation instead of using all of $\mathbf{u}_i$, using only the first few $\mathbf{u}_i$ allows us to obtain an approximate solution for neutron spectra. In other words, this method enables dimensionality reduction. Hereafter, the number of the vectors $\mathbf{u}_i$ used will be referred to as the *subspace dimension*.

2.2 Determining the Coefficients of a Linear Regression Function

Next, we created a regression model to determine the expansion coefficient for the neutron spectrum from the input parameter vector $\mathbf{p}$. First, we express the neutron flux ϕ using the $\hat{\mathbf{U}}$ matrix and the expansion coefficient vector $\mathbf{a}$ as $\phi = \hat{\mathbf{U}}\mathbf{a}$. To create a regression model to predict the expansion coefficients a_j , which is redefined as d for simplicity, which is the j th element of $\mathbf{a}$ using polynomial function f_i as follows:

$$d \simeq \sum_{i=1}^{n_i} b_i f_i(\mathbf{p}). \quad (5)$$

We could obtain n_r sets of $\mathbf{p}$ vector and d . Using them to determine b_i , we defined a matrix $\mathbf{X}$, vectors $\mathbf{b}$, and $\mathbf{d}$ as follows:

$$\mathbf{X} = \begin{bmatrix} f_1(\mathbf{p}_1) & \cdots & f_{n_i}(\mathbf{p}_1) \\ \vdots & \ddots & \vdots \\ f_1(\mathbf{p}_{n_r}) & \cdots & f_{n_i}(\mathbf{p}_{n_r}) \end{bmatrix}, \quad (6)$$

$$\mathbf{b} = (b_1 \quad b_2 \quad \cdots \quad b_{n_i})^T, \quad (7)$$

$$\mathbf{d} = (d_1 \quad d_2 \quad \cdots \quad d_{n_r})^T. \quad (8)$$

Next, we used the least-square method to obtain vector $\mathbf{b}$; i.e. multiplying the pseudoinverse of $\mathbf{X}$ to $\mathbf{d}$.

$$\mathbf{b} = \mathbf{X}^\dagger \mathbf{d}. \quad (9)$$

3. NUMERICAL TEST

Based on the theory described in the preceding section, we created a regression model with using the training data. At each burnup step of burnup calculations, we predicted expansion coefficients of POD components using the created regression model from given input parameters, and based on this, we calculated the multi-group neutron flux. Using this, we calculated the reaction rate to perform burnup calculations to the next step. As a result of it, the prediction accuracy of the nuclide inventories was evaluated.

3.1. Computational Conditions

The gas-cooled reactor used for this calculation is Japan's HTTR [3]. Its core consists of hexagonal graphite blocks with a side length of 36 cm. The fuel rods are formed by melting and solidifying multiple TRISO particles, approximately 1 mm in diameter, into the graphite matrix inside the graphite cladding tube. Here, TRISO particles are UO_2 fuel coated with low-density PyC, high-

density PyC, and SiC. Fuel rods with this structure are inserted into several channels formed in a graphite block, with the graphite matrix and graphite block functioning as moderators. Helium gas flows within the gaps inside each channel, serving as the coolant.

We used the CBZ code [4] to perform burnup calculations targeted HTTR with input parameters including U-235 enrichment, fuel compact cell pitch, TRISO packing fraction, linear power, and fuel temperature. Snapshot matrices (361×39) of neutron energy spectra for 361 groups [5] at 39 burnup points within the range of 0 to 90 GWD/t were extracted.

Specifically, within the ranges shown in Table I, all combinations of the maximum, median, and minimum values for each parameter were used to obtain a total of 243 sets of training data. In addition, we used the stratified sampling method also. By obtaining 100 random neutron energy spectra from each of four partitions of the parameter sampling range, a total of 343 sets of 361×39 neutron energy spectra were acquired. These were used to create a 361×13377 snapshot matrix.

The total number of regression functions n_i is determined by the maximum order of the regression polynomial and the number of parameters. When the maximum order of the regression polynomial is n , all combinations with order less than or equal to n become the number of functions. Therefore, by using the number of input parameters n_c , the number of functions n_i is expressed as $n_i = \frac{(n+n_c)!}{n_c!n!}$. The number of functions in the linear regression function determined by this equation is shown in Table II.

The accuracy of the regression model was verified using 120 random test conditions. Specifically, for each burnup step in the burnup calculation using CBZ, the method of calculating the neutron energy spectrum by solving the neutron transport equation was replaced with one that predicts it using the regression model, and the nuclide inventories was calculated at each burnup point.

Table I. Sampling range of parameters.

Parameter	Range
U-235 enrichment (wt%)	3.4 - 9.9
Fuel compact pitch (cm)	5.05 - 5.75
TRISO packing fraction (-)	0.30 - 0.60
Linear power (W/cm)	40 - 200
Fuel temperature (K)	900 - 1500

Table II. Number of regression functions with the maximum order of the polynomial.

Maximum order of the polynomial	Number of regression functions
3	84
4	210
5	462
6	924

3.2. Result

First, the relative error between the nuclide inventories predicted by the regression model and that calculated by the CBZ code was quantified. The maximum order of the polynomial was fixed at 4, while the subspace dimension was varied to 4, 6, 8, 10, 12, and 14.

The RMS and maximum errors of number densities of Cs-134, U-235, Pu-239, and Am-241 at each burnup point are shown in Figures 1 to 4. Figures 1 and 2 show comparisons where the maximum order of the regression polynomial is fixed at 4, while varying the subspace dimension used. Figures 3 and 4 show comparisons where the subspace dimension used is fixed at 14, while varying the maximum order of the regression polynomial.

When the maximum order of the polynomial and the subspace dimension were set to 4 and 14, respectively, the RMS error was below 5% in most cases, and the maximum error rarely exceeded 10%. Therefore, it is considered that sufficient accuracy was achieved.

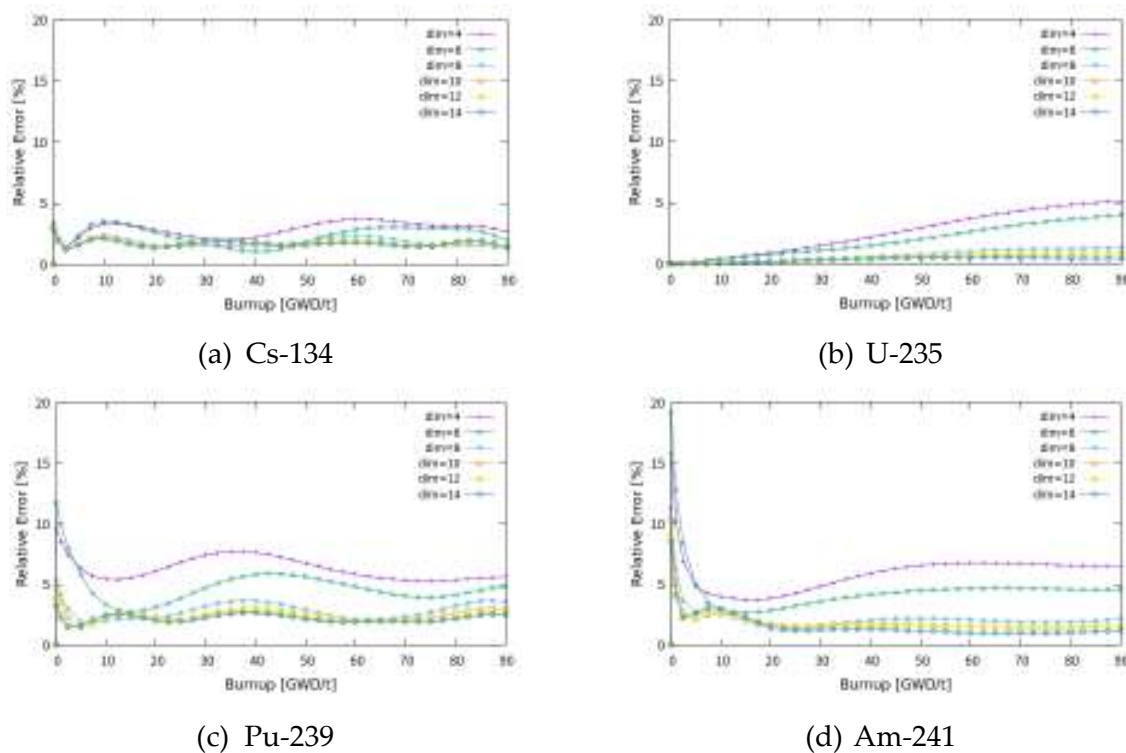
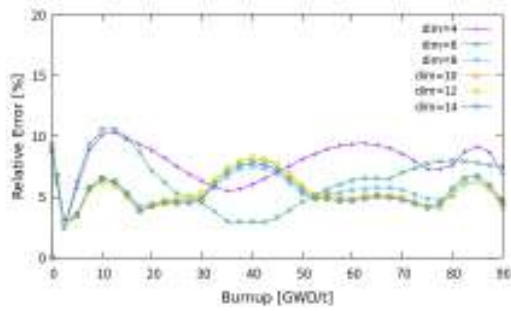
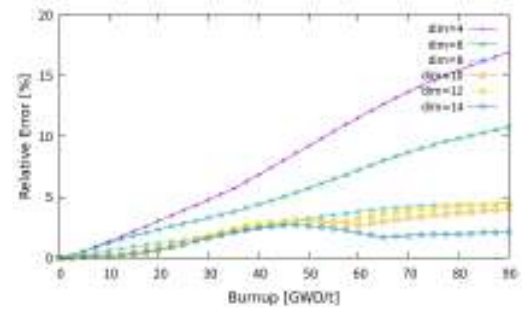


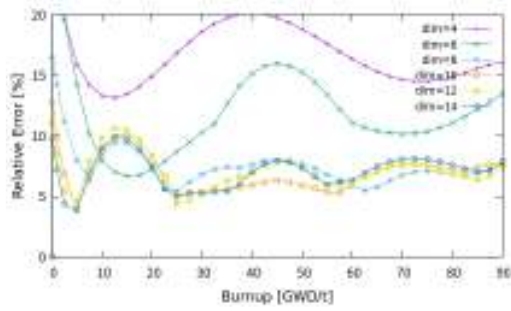
Figure1. RMS error in inventory per burnup for 120 samples with different subspace dimension.



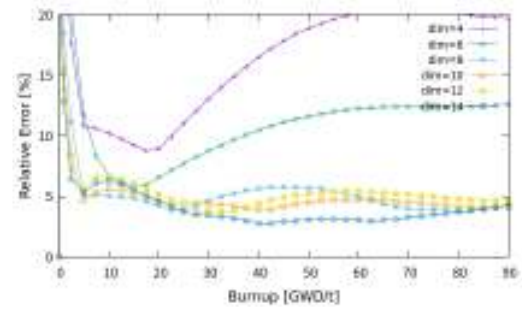
(a) Cs-134



(b) U-235

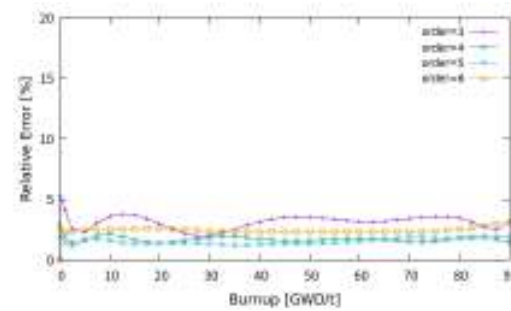


(c) Pu-239

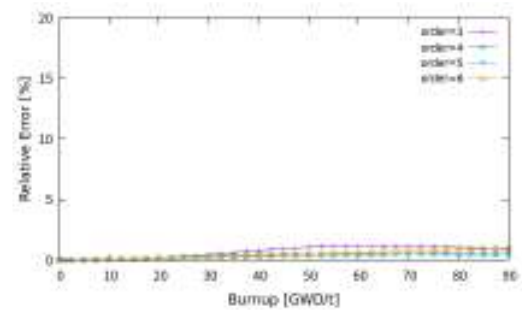


(d) Am-241

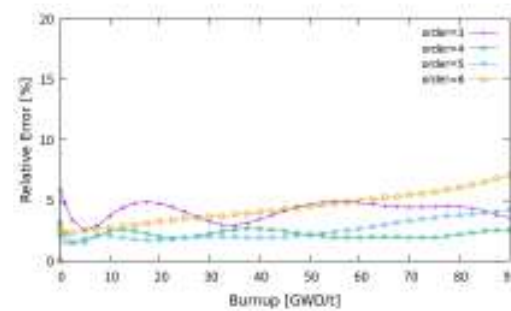
Figure2. Maximum error in inventory per burnup for 120 samples with different subspace dimension.



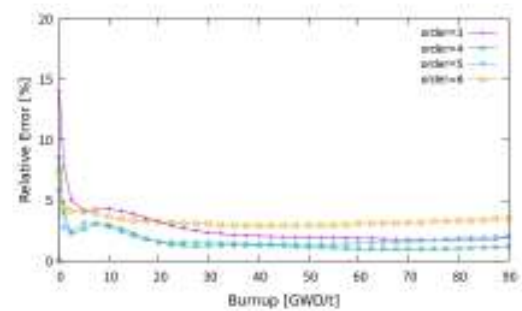
(a) Cs-134



(b) U-235



(c) Pu-239



(d) Am-241

Figure3. RMS error in inventory per burnup for 120 samples with different maximum order of the regression polynomial.

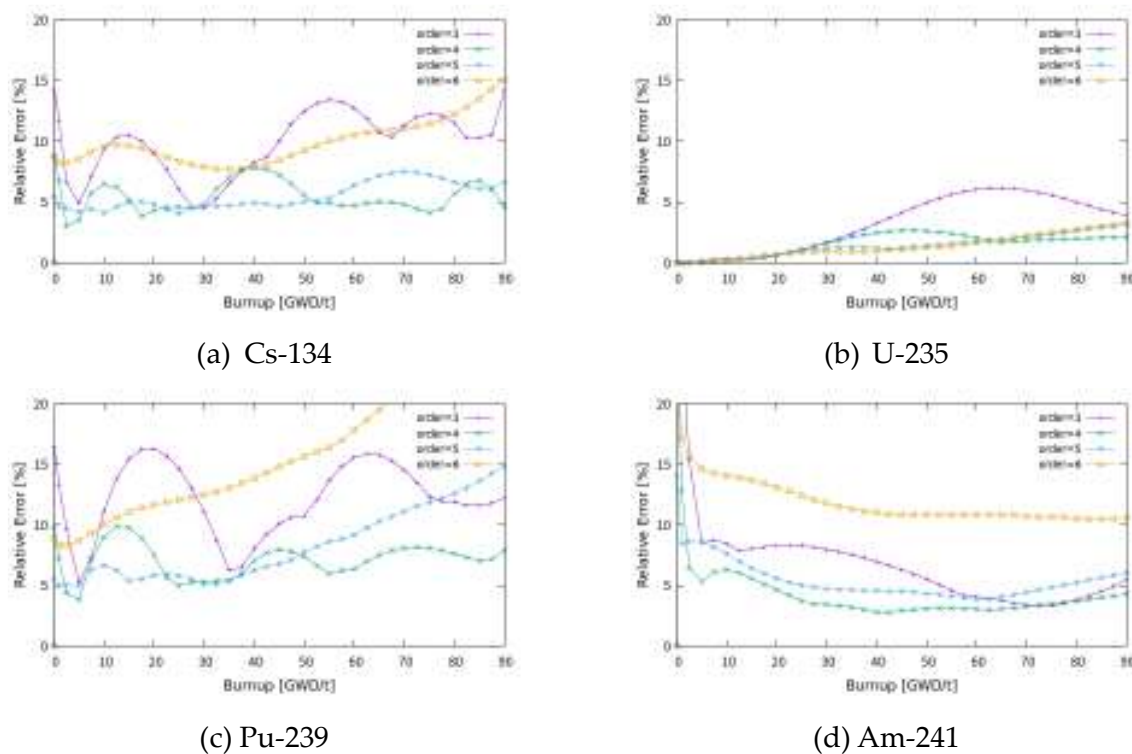


Figure 4. Maximum error in inventory per burnup for 120 samples with different maximum order of the regression polynomial.

3.3. Discussion

We considered that two main factors were causing the errors in neutron flux energy spectrum reconstruction. The first is error due to the shape of the adopted regression function. The second is error due to dimensionality reduction caused by limiting the number of subspace dimension used. In other words, the accuracy of this regression model depends on the maximum order of the polynomial and subspace dimension used.

Regarding the maximum polynomial order, while a low order like three may result in poor regression accuracy due to an insufficiently descriptive regression function shape, accuracy also declined when set to 5th or 6th order. This is likely because a higher-order regression function amplifies small errors.

As shown in the results, increasing the subspace dimension generally improves accuracy. However, no clear difference was observed between the results when using dimensions of 10, 12, and 14. This is likely because the importance of subspace dimension decreases as their dimension increases; thus, using subspace dimension of a certain dimension or higher yields only minor changes in accuracy.

Furthermore, changing the sampling method for the training data could be a way to further improve the accuracy of this regression model. For example, this study used a total of 243 cases as training data, comprising all combinations of the maximum, median, and minimum values for the five input parameters. Additionally, 100 random cases sampled using the stratified sampling method were prepared. This approach accounts for the possibility that sample data set causing extreme increases in error may exist not only near the maximum, median, and minimum values of the five parameters. However, overusing data sampled by the stratified sampling method as training data can diminish the influence of training data sampled at mesh boundaries on the regression function. Therefore, we believe there is a room for further research regarding the appropriate amount of stratified-sampled

data to utilize.

4. CONCLUSIONS

As a result of this study, by employing dimensionality reduction using POD and the least-square method, we successfully developed a regression model for the neutron energy spectrum in high-temperature gas-cooled reactor HTTR. This model reconstructs the neutron energy spectrum within the fuel compact for any burnup and operating conditions, given multiple input parameters such as U-235 enrichment and fuel temperature. Furthermore, it was confirmed that by substituting random test conditions into this model and performing calculations, the nuclide inventory could be predicted with an accuracy of less than few-percent errors. Finally, the present model is formulated at the fuel compact level and intended for fast burnup calculations under prescribed irradiation conditions, and its extension to explicitly incorporate lattice- or core-level environmental effects is identified as an important topic for future work.

ACKNOWLEDGMENTS

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